

Utsav Munendra

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WORK EXPERIENCE

Meta, Software Engineer Jan 2022 - Present
Virtual Reality Firmware Burlingame, CA

- Brought up boards, sensors and interfaces for MCUs in Meta Quest headsets and Meta Touch controllers.
- Brought up USB Mass Storage end-to-end solution, from AOSP filesystem integration to flash sensor drivers. Committed bugfixes to LittleFS and Zephyr.
- Designed sensor driver framework and HAL implementations for efficient sensor sampling from Qualcomm Hexagon DSP while targeting reusability with other MCUs.
- Improved compute load and modularity of sensor data streaming APIs via Linux kernel driver optimizations.
- Led in-market support for camera firmware involving architecture improvements, agentic bug-resolution, AI-assisted data-analysis along with collaboration with Qualcomm on CamX.
- Led JMD as the sole partner SWE for Meta Quest Controller Rechargeable Batteries, and designed its interface and protocol with OEM controllers.
- Assisted various factory builds via on-site debug and bringup support along with tooling implementation.

Green Hills Software, Software Engineering Intern Summer 2021, Summer 2022
MULTI Debugger & Advanced Products Group Santa Barbara, CA

- Extended the Rust compiler and standard library to natively support new Green Hills' Integrity RTOS targets. Proved support by hosting a web server on targets backed by prevalent async runtime Rust crates.
- Increased performance of GHS Embedded Systems Debugger by making IPC more efficient.

Verkada, Software Engineering Intern Winter 2021, Spring 2021
Cameras Core OS San Mateo, CA

- Assisted in board bringup for CD62 Dome camera model and created the test rack for the new hardware.
- Brought up OS service in Go for system encrypted volume setup and filesystem management.

EDUCATION

University of California, Los Angeles Sept 2018 - Dec 2021

B.S. in Computer Science, Henry Samueli School of Engineering and Applied Science

GPA: 3.96, *summa cum laude*

Reader and undergraduate TA for CS33: Computer Organization (Systems Architecture)

PROJECTS

Open-Source Contributions to Zephyr RTOS 2025

Approved contributor for Zephyr Project, merged patches for various flash, MSPI driver and compatibility layer fixes.

WebAssembly Runtime (WAMR) for ARM Mbed 2020

Allows existing WebAssembly modules to run on ARM Mbed boards and the web alike for increased portability for IoT applications.

SKILLS

Languages C/C++, Python, Rust, Go

RTOS Zephyr, RTX5, Qualcomm QuRT

Tools GDB, CMake, Makefile, BitBake, Logic Analyzer, Oscilloscope